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# Detection of Low-Altitude Unmanned Vehicle and Flying Bird by Improved YOLOv7

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## ABSTRACT

With the rapid development of low-altitude economy and unmanned aerial vehicle (UAV) technologies, reliable UAV surveillance has become increasingly important for aviation safety and low-altitude airspace management. However, accurate recognition of UAVs in optical imagery remains challenging due to small target sizes, motion blur, complex backgrounds, and the visual similarity between UAVs and flying birds. To address these challenges, this paper proposes an improved YOLOv7-based framework for low-altitude UAV and bird recognition. The proposed framework adopts MobileNetV3 as a lightweight backbone to reduce computational complexity while maintaining efficient feature extraction capability. To improve small-target detection performance, a modified BiFPN and SPD-Conv based small-object enhancement module are introduced to enhance multiscale feature fusion and preserve fine-grained spatial information. Furthermore, a trajectory-aware recognition strategy based on a simplified Global Tracking Transformer (GTR) is incorporated to exploit temporal motion characteristics and improve UAV–bird discrimination in dynamic scenes. In addition, a dedicated UAV–Bird dataset containing realistic low-altitude surveillance scenarios is constructed for training and evaluation. Extensive experiments are conducted on multiple public datasets, including MAV-VID, Anti-UAV RGB, and an open source Drone vs. Birds Dataset from Kaggle, as well as the proposed dataset. The results demonstrate that the proposed framework achieves competitive detection performance while maintaining real-time inference capability. On the self-built dataset, the proposed method improves mAP@0.5 from 0.927 to 0.933 and F1 score from 0.903 to 0.914 compared with the original YOLOv7 while increasing the inference speed from 33.6 to 55.6 FPS. These results indicate that the proposed framework effectively balances detection accuracy and computational efficiency, making it suitable for practical low-altitude surveillance applications.

## 1 | Introduction

With the rapid development of low-altitude economy and unmanned aerial vehicle (UAV) technologies, UAVs have been increasingly deployed in logistics, aerial photography, environmental monitoring, agriculture, and emergency response applications. However, the widespread utilization of UAVs has also introduced significant safety concerns, particularly in sensitive airspace environments such as airports, urban low-altitude corridors, and restricted regions. In recent years, multiple incidents involving unauthorized UAV intrusions have threatened the safety of civil aviation operations and public infrastructure. Consequently, accurate and real-time UAV surveillance has become an important research topic in intelligent airspace security systems.

Currently, drone detection predominantly relies on radar systems [1]. Although radar is effective in detecting airborne targets, it remains vulnerable to interference from other objects like flying birds and electromagnetic environmental factors.

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Additionally, radar alone may not accurately identify and distinguish specific targets, hindering precise UAV detection. In contrast, vision-based target identification bypasses the limitations associated with radar detection. The process involves radar detecting the target azimuth, followed by the introduction of optoelectronic equipment to precisely identify the target from captured images, completing the UAV target recognition process [2].

Benefiting from the rapid advancement of deep learning and computer vision technologies, convolutional neural network (CNN)-based object detection algorithms have achieved remarkable success in aerial target detection tasks. Existing deep-learning-based detectors can generally be divided into two categories: two-stage detectors and one-stage detectors. Two-stage detectors—represented by R-CNN [3], Faster R-CNN [4], and Mask R-CNN [5]—first generate region proposals and then perform feature extraction and classification. These methods usually achieve high detection accuracy but suffer from relatively high computational complexity and limited inference speed. In contrast, one-stage detectors, such as SSD [6] and the YOLO [7] series, directly perform localization and classification in a unified framework, thereby achieving significantly higher detection efficiency and better suitability for real-time applications.

Among existing one-stage detectors, YOLOv7 [8] has demonstrated an excellent balance between detection accuracy and inference speed. Owing to its efficient feature extraction architecture and optimized training strategy, YOLOv7 has become one of the most representative real-time object detection frameworks. However, directly applying YOLOv7 to low-altitude UAV and bird recognition scenarios still faces several critical challenges [9].

First, low-altitude aerial targets usually occupy extremely limited pixel regions in images [10]. In many long-distance surveillance scenarios, the target size may be smaller than $32 \times 32$ pixels, leading to significant feature loss during down-sampling operations. Existing detection frameworks often exhibit substantial performance degradation under such small-target conditions. Second, aerial targets frequently suffer from motion blur caused by rapid movement and camera jitter, resulting in incomplete appearance features and unstable spatial representations [11]. Third, UAVs and birds often exhibit highly similar visual characteristics in optical imagery, especially under complex backgrounds and adverse illumination conditions. Appearance-based detectors relying solely on single-frame information therefore tend to generate false alarms and target confusion. In addition, target occlusion and multiscale variations further increase the difficulty of robust detection and recognition in practical low-altitude environments [12].

To address the above challenges, this paper proposes an improved YOLOv7-based UAV and flying bird detection framework for low-altitude aerial surveillance scenarios. First, a lightweight backbone based on MobileNetV3 is introduced to reduce computational complexity while maintaining efficient feature extraction capability. Second, a modified multiscale feature fusion structure integrating BiFPN-style aggregation and SPD-Conv is imported to enhance the preservation of fine-grained spatial features for small-scale targets. Third, a trajectory-aware recognition strategy is incorporated to exploit temporal motion characteristics for UAV and bird discrimination in dynamic scenes. By combining appearance information and motion-aware analysis, the proposed framework improves recognition robustness under complex aerial environments.

Moreover, a UAV–bird dataset containing realistic low-altitude aerial scenes is constructed to evaluate the proposed framework. The dataset includes various target scales, motion states, background complexities, and bird interference scenarios, providing a practical benchmark for UAV–bird discrimination tasks.

The main contributions of this work are summarized as follows:

- A lightweight UAV–bird detection framework based on YOLOv7 is proposed for low-altitude aerial surveillance scenarios, achieving a balance between detection accuracy and real-time performance.
- A modified BiFPN and SPD-Conv based small-object enhancement module is designed to enhance the representation ability for small-scale aerial targets.
- A trajectory-aware recognition strategy is introduced to incorporate temporal motion characteristics into UAV and bird discrimination, improving robustness in dynamic scenes.
- Proposed a dataset which is similar to a common application scenario. It can be used to train models for actual use, and with the ability to distinguish drones from flying birds. This dataset may provide important reference significance for follow-up research.

The remainder of this paper is organized as follows: Chapter 2 reviews related works on UAV detection and aerial target recognition; Chapter 3 presents the proposed methodology in detail; Chapter 4 describes the experimental settings, datasets, and evaluation metrics; Chapter 5 discusses the experimental results and ablation studies; and finally, Chapter 6 concludes this paper and outlines future research directions.

## 2 | State of the Art

Because the introduction of deep learning technology for the detection and recognition of drones began in 2016 [13, 14], numerous UAV detection algorithms based on CNNs have been proposed. Particularly from 2017 onwards, these algorithms encompass various techniques for classifying UAVs and birds [15]. Within this research domain, one approach involves directly enhancing image recognition algorithms to better suit the current application scenarios. This entails refining existing algorithms to improve their performance specifically for UAV recognition. Researchers have explored modifications and optimizations to CNN architectures, network parameters, and training methodologies to achieve more accurate and efficient UAV classification results. Another avenue of research involves combining the CNN-based algorithms



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with other complementary techniques to enhance the overall detection effectiveness of the target. For instance, motion tracking methods can be incorporated to track the UAVs [16], providing temporal context and aiding in accurate recognition. Additionally, multiframe fusion techniques [17] can be employed to fuse information from multiple frames or images, allowing for a more comprehensive understanding of the target and improving recognition accuracy, especially in challenging scenarios with occlusions or complex backgrounds. These approaches demonstrate the diversification of UAV recognition research, encompassing both direct algorithmic improvements and integration of complementary methods to achieve higher accuracy on detection. The field continues to evolve as researchers explore innovative techniques and refine existing approaches to improve the performance of UAV detection algorithms.

Judging from specific related studies. Sommer et al. [18] proposed combining region of interest selection with Faster R-CNN in the Drone vs. Birds Challenge at the 14th International Conference on Advanced Video and Signal Based Surveillance in 2017 (AVSS 2017). The feature-extraction backbone network introduces confidence weights in the network classification layer, enhances the image extraction efficiency of corresponding targets, and achieves the best results for UAV and bird identification in the current challenge. In the same challenge, Aker et al. [19] introduced the YOLO algorithm into application; they used the K-means method to filter the confidence interval of the scale box of the dataset, select the most suitable scale for UAV and bird detection, and then set the corresponding scale in YOLOv2. This method has strong robustness for identifying different target types and has a much higher rate in recognition than the same algorithm in target detection. Sommer et al. [20] proposed a method of first detection and then classification, separating the flying target from the background by the segmentation method. It extracts objects that do not belong to the background and uses the area detection network to determine their coordinates in the image. Then VGG combined with the constructed classification network is used to train the UAV and bird classifier obtained to realize the classification operation of UAV and flying bird. It obtains satisfactory results relatively. In 2018, Carrio et al. [21] applied depth maps to the detection of drones and used AirSim software to generate virtual targets for the expansion of the depth of the drone dataset, realizing the combination of real data and virtual data. At the third Drone vs. Birds Challenge in 2019, Magoulianitis et al. [22] proposed the use of super-resolution method for UAV detection, by amplifying local features, reconstructing the corresponding learning network, and fine-tuning in training to adapt to UAV detection and recognition tasks. Finally, it builds an efficient and low-cost network structure. Nalamati et al. [23] obtained the best detection results by using the Faster-RCNN algorithm with ResNet-101 as the main network by applying and comparing the existing CNNs. The following year, in the 2020 Drone vs. Birds challenge, Coluccia et al. [24] used Cascade R-CNN [25] to achieve optimal recognition results. In the 2021 event, Dadboud et al. [26] benefited from the development of YOLO series algorithms. They applied the updated YOLOv5 to achieve the ability to identify targets with accuracy at a high rate.

At the same year as 2021, an improved Resnet50 as a feature extraction module was utilized by Ashraf et al. [27]. Their goal is detecting drones in video frames, which are captured by other flying objects. To deal with some problems such as small target detection, complex background, and abrupt motion of targets etc. They do some work based on special and spatiotemporal. They use all four scales of Resnet50 to achieve initially feature extraction processing, then extracted features are concatenated and resized to gather multiscale information which can enhance the feature extraction ability of small targets. They integrated channel attention with a pixel weight-based attention mechanism to create a multiattention system, resulting in a dual CNN for extracting and classifying targets in videos with dynamic backgrounds. Although this method performs well in these environments, it still faces challenges in distinguishing targets from similar background.

In 2022, Akyon et al. [28] tackled the issue of target detection in real-world videos, proposing enhancements for the detection and tracking of continuous targets. Utilizing the Drone vs. Birds dataset, they combined YOLOv5 with Kalman filtering to detect and track moving targets while delegating the task of target classification to a subsequent sequence classification network. The authors presented two distinct classification approaches: the 3D CNN R(2 + 1)D and a Resnet18-based algorithm augmented with an added feature extraction neck network. Experimental results indicated that R(2 + 1)D achieved the highest overall F1 score of 89.2 for detection, whereas LSTM significantly enhanced the feature extraction capabilities of Resnet in video sequences.

Ihekoronye et al. [29] investigated the detection performance of drones and other flying targets under clear, cloudy, and nighttime conditions. They introduced the KITYOLO network, based on YOLOv5, and compared its performance to the original YOLOv5 algorithm, finding accuracy improvements of 11.2%, 47.4%, and 32.4% under these respective conditions. Liu et al. [30] proposed MUSAK, a multiscale spatial motion method for aerial target detection. This technique combines static image data with motion feature parameters, extracting polar geometric features to use edge pixels for matching the same target from different angles. The method employs a GRU structure for feature classification, enabling continuous target detection even when surface feature information is missing. However, as MUSAK relies on changes in appearance features, it is constrained by feature scale variations and cannot fully match targets with significant missing appearance features.

At the 2023 ICASSP conference, more SOTAs for drone and bird detection were presented. Kim et al. [31] utilized YOLOv8 combined with multiscale image fusion to detect high-speed drones, participating in the sixth WOSDETC challenge. Although they achieved some improvements in detecting faster drones, the overall accuracy was still surpassed by Mistry et al. [32] who employed a boundary tracker with YOLOv7 to detect drones and birds. Mistry's method achieved over 90% accuracy in multiple video sequences with a uniform background. However, the accuracy dropped significantly in complex backgrounds or when the targets were small, sometimes resulting in worse outcomes compared to earlier targeted improvements.

Furthermore, there are more methods for the detection of drones and birds that have been raised in recent days. For low altitude background, Ma et al. [33] modified the SimAM attention module



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and proposed a fusion block based on the normalized Wasserstein distance and achieved significant results for low altitude UAV detection. Zhu et al. [34] applied a modified YOLOv9 using GAM attention mechanism and CAM feature enhancement module and introduced a dynamic convolution called AKConv which can focus on more relevant feature regions of input characteristics. They proposed a unique method for multiscenario detection of UAVs.

As for the validation method for various modified algorithms, the introduction of an evaluation index for evaluating the use of deep neural networks in UAV detection and tracking, proposed by Isaac-Medina [35] at the AVSS conference in 2020, marks a significant advancement in the field of computer vision and aerial surveillance. Prior to this development, there might not have been a standardized and widely accepted benchmark to measure the performance of algorithms specifically designed for UAV detection and tracking. The evaluation index serves as a valuable tool for researchers, practitioners, and developers in the domain of UAV detection and tracking. It allows them to objectively assess the effectiveness, accuracy, and efficiency of different deep neural network-based algorithms for this critical application.

Different from existing attention-based YOLO improvements, the proposed method focuses on lightweight multiscale-feature preservation and motion-aware target discrimination under low-altitude aerial scenarios.

## 3 | Proposed Approaches

In this study, we use the YOLOv7 network as the basement, and specific enhancements will be implemented to tailor the algorithm for improved detection of drones and birds. YOLOv7 was introduced in 2022 by Chien-Yao Wang and Alexey Bochkovskiy [8], who are also authors of YOLOv4 [36], it represents a significant evolution. In contrast to YOLOv4, YOLOv7 introduces substantial changes, including modifications to the backbone network, convolution methods, and classifier. These alterations result in faster and more accurate detection and recognition of targets.

Figure 1 shows the structure of YOLOv7; the network of YOLOv7 is divided into three parts, separately called backbone, neck, and head. In the backbone structure, ELAN is the key section. It is used to extract features with in-depth information of targets in optical images. Neck extracts more information from forward feature matrix, and head part achieves the classification of various targets.

However, YOLOv7 still encounters the ongoing challenge of effectively detecting small targets, which has been a persistent issue throughout the YOLO series of networks. To overcome this drawback, we propose the utilization of a scale-enhanced method to enhance the features of feature-blurred targets.

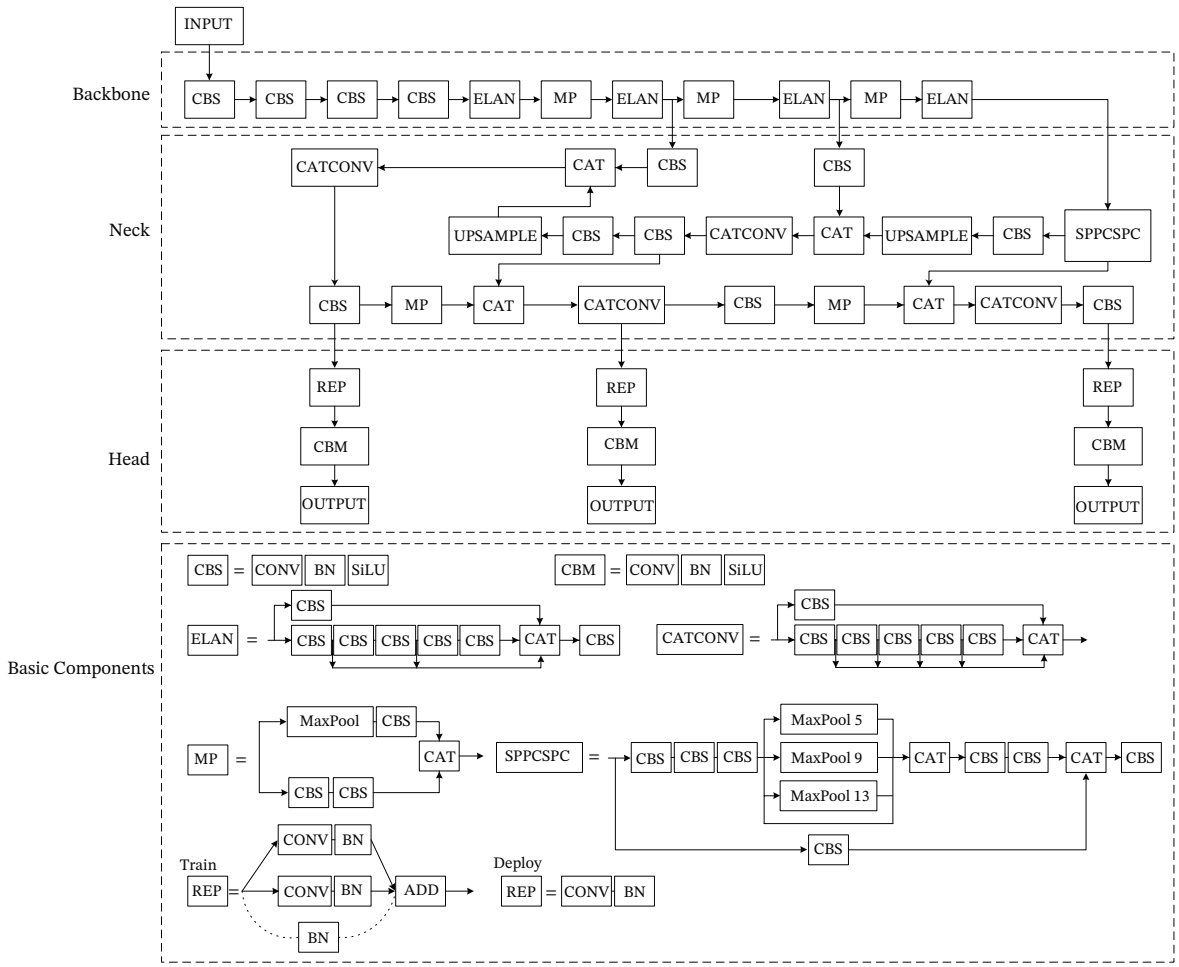



**FIGURE 1** | Structure of YOLOv7.



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### 3.1 | Lightweight Backbone

Although YOLOv7 achieves excellent detection accuracy, its original backbone network introduces relatively high computational complexity and memory consumption, which limits deployment on embedded systems and consumer-grade edge-computing platforms. To improve inference speed while maintaining acceptable recognition performance, MobileNetV3 [37] is adopted to replace the original YOLOv7 backbone.

MobileNetV3 utilizes depth wise separable convolution and lightweight inverted residual bottleneck structures to significantly reduce the number of parameters and floating-point operations (FLOPs). Compared with standard convolution, depth wise separable convolution decomposes feature extraction into depth wise convolution and pointwise convolution.

Furthermore, MobileNetV3 integrates squeeze-and-excitation (SE) attention mechanisms and hard-swish activation functions, enabling efficient feature recalibration under lightweight constraints. To adapt the detector to the modified backbone, the feature aggregation layers and detection heads of YOLOv7 are correspondingly adjusted. The comparison between the original YOLOv7 backbone and the proposed lightweight backbone is shown in Figure 2.

By replacing the original backbone with MobileNetV3, the proposed framework substantially reduces model complexity and improves inference speed, making it more suitable for real-time UAV surveillance applications.

### 3.2 | Small Objects Enhance Module

In this research, due to the actual targets that are captured by optoelectronic devices usually have small scales. Our primary objective is to enhance the network's capability to detect microtargets [38], which are defined based on both relative and absolute scale. Relative scale refers to objects whose bounding box area is between 0.08% and 0.58% of the image area, whereas absolute scale considers objects with pixel dimensions smaller than $32 \times 32$ defined as microtargets. Identifying and recognizing such small objects in pictures can be challenging even for humans, requiring extra attention and specialized techniques. To overcome this challenge, we propose a module that offers a clear solution for small target detection.

As common means, the attention mechanism [39, 40] serves as a widely utilized method for assigning weights to different regions of an image. It enables the detector to focus on key regions, resulting in more targeted object detection. Additionally, the attention mechanism can also enhance performance in detecting dynamic targets that may vary in appearance over time. Despite its advantages, the attention mechanism has certain limitations. One drawback is that it can increase the complexity of the network, making optimization more challenging. Furthermore, due to substantial uncertainties, accurately predicting whether the attention mechanism will positively impact the algorithm's results becomes difficult. To overcome these limitations, we introduce a modified Feature Pyramid Network (FPN) [41]. By integrating FPN, we aim to enhance the quality of features extracted from small targets. FPN has been proven effective in addressing scale variations and capturing features at different resolutions. This integration serves to improve the performance of vision-based recognition methods, particularly in detecting small targets. By combining the attention mechanism and the modified FPN feature extraction backbone, our proposed module significantly enhances the network's ability to detect microtargets. This approach enables the detector to focus on crucial image regions while simultaneously capturing multiscale features, leading to improved recognition performance for small objects.

Our module is modified based on PANet [42], and borrowing the ideas of BiFPN [43] construction to improve. We build modules modeled on PANet and BiFPN, based on Simplified PANet, add a hop connection to the upward path, shortening the path from low-level information to high-level information, so that more low-level feature information is retained. The modified module is added in the neck of YOLOv7 to enhance the ability to process feature information. Figure 3 shows the structure of our modified-FPN module.

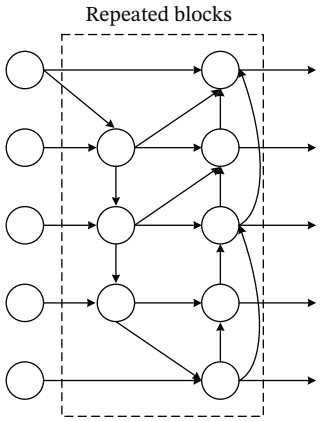



**FIGURE 3** | Structure of modified-FPN module. This structure is added to the neck of YOLOv7. More low-level features can be left so that some small objects' features, which may give up, are preserved.

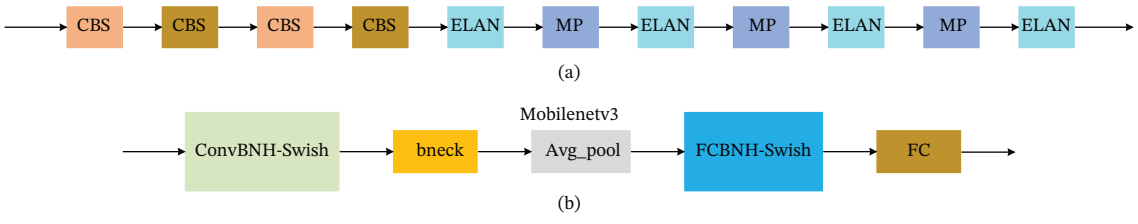



**FIGURE 2** | Comparison of the structure of (a) YOLOv7 backbone and (b) MobileNetV3.



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Besides, repeated strided convolution and pooling operations in conventional CNNs inevitably cause spatial information loss, particularly for tiny objects. To further preserve fine-grained appearance information, SPD-Conv [44] is introduced into the neck network. SPD-Conv transforms spatial-domain information into channel-domain representations through space-to-depth transformation, thereby avoiding direct feature elimination during down-sampling.

The transformation process can be formulated as follows:

$$X_{SPD} \in \mathbb{R}^{(H/s)\times(W/s)\times Cs^2}, \tag{1}$$

where $H$ and $W$ denote spatial dimensions, $C$ represents channel number, and $s$ denotes stride size.

Compared with conventional down-sampling operations, SPD-Conv preserves more appearance details for small targets by converting spatial features into channel representations. In the proposed framework, SPD-Conv is inserted between the backbone network and neck structure, enabling more discriminative feature information to be retained before feature fusion. This operation significantly improves detection sensitivity for distant UAVs and blurred bird targets.

### 3.3 | Track to Recognition

For moving objects in video or multiframe images, object tracking algorithms can also be used as an important means of object recognition [45]. It can realize continuous tracking after a vision-based identification of the target, without spending a lot of computing resources for repeated analysis every frame and can achieve rapid detection and identification when the target is briefly lost and then reappeared. In the classification and identification of UAVs and flying birds, the motion characteristics of UAVs and flying birds are quite different, so the method of global trajectory feature extraction is considered to analyze and predict the target trajectory and realize tracking. Global Tracking Transformers (GTRs) [46] is an end-to-end tracking method based on video with 32 frames. It follows the idea of detecting first and then tracking, which is short for tracking by detection.

The method of predicting the motion trajectory of the target by extracting the motion characteristics is introduced to realize the motion tracking of the target, which has high requirements for the motion characteristics of the target. The difference in the motion attitude of drones and birds in the air is just suitable for this kind of motion trajectory feature extraction and classification scene. Analyze with video or multiframe image datasets of relevant targets. In two-dimensional image space, aerial targets are projected as planar objects [47]. During motion, radial displacement mainly causes scale variation, whereas lateral motion leads to obvious positional displacement between adjacent frames. Consequently, UAVs and birds exhibit distinct temporal motion signatures in image sequences. Figure 4 illustrates the periodic trajectory characteristics of flying birds, whereas Figure 5 presents typical UAV motion trajectories captured by an optical camera system.



**FIGURE 4** | Trajectories of flying birds.

Given that the detection result set is at frame $t$, To associate detections across temporal frames, the trajectory matching criterion is defined based on the intersection-over-union (IoU) metric:

$$B^t = \left\{b_1^t, b_2^t, \cdots, b_i^t\right\}. \tag{2}$$

$$\widehat{\alpha}_k^t = \begin{cases} \text{ØØØØØØØØ}, if\tau_k^t = \text{ØØØØØØØØ} or \max_i IOU\left(b_i^t, \tau_k^t\right) < 0.5 \\ argmax_i IOU\left(b_i^t, \tau_k^t\right), otherwise \end{cases}, \tag{3}$$

where $\widehat{\alpha}_k^t$ is used as the object matched with ground-truth trajectory $\tau_k^t$ according to the equation.

Our goal is to identify the target type by tracking the target's trajectory and analyzing its motion characteristics. Therefore, we exploit trajectory dynamics and motion patterns for discriminative modeling. The GTR framework is adapted for the proposed scenario. Specifically, the original GTR architecture is simplified by removing its internal appearance feature extraction branch, which reduces computational cost and improves efficiency. In the proposed design, appearance features are extracted from an external detection network and serve as complementary visual cues. These appearance features are fused with trajectory representations generated by the motion modeling module, enabling joint utilization of motion and visual information. This design improves discriminative performance while keeping the model lightweight. Because aerial objects such as UAVs and birds typically exhibit continuous motion with stable short-term dynamics, motion cues provide strong discriminative power. Therefore, the proposed method focuses on motion-aware trajectory modeling, using appearance features only as auxiliary information for robust recognition.

### 3.4 | Overall Architecture

The architecture of the proposed UAV–bird recognition framework is illustrated in Figure 6. The framework follows a detection–tracking–recognition paradigm and consists of three main components: a lightweight detection branch, a temporal tracking branch, and a motion-aware recognition branch.

First, input image sequences are processed by an improved YOLOv7 detector. To reduce computational complexity and



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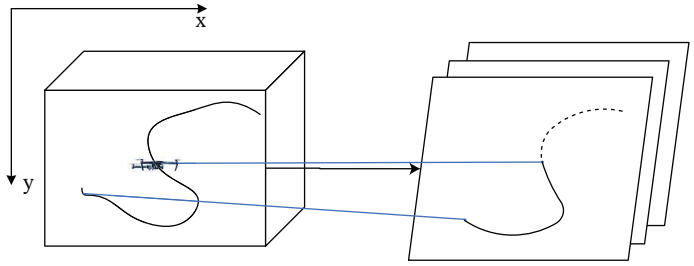



**FIGURE 5** | Motion trajectory of the drone in camera.

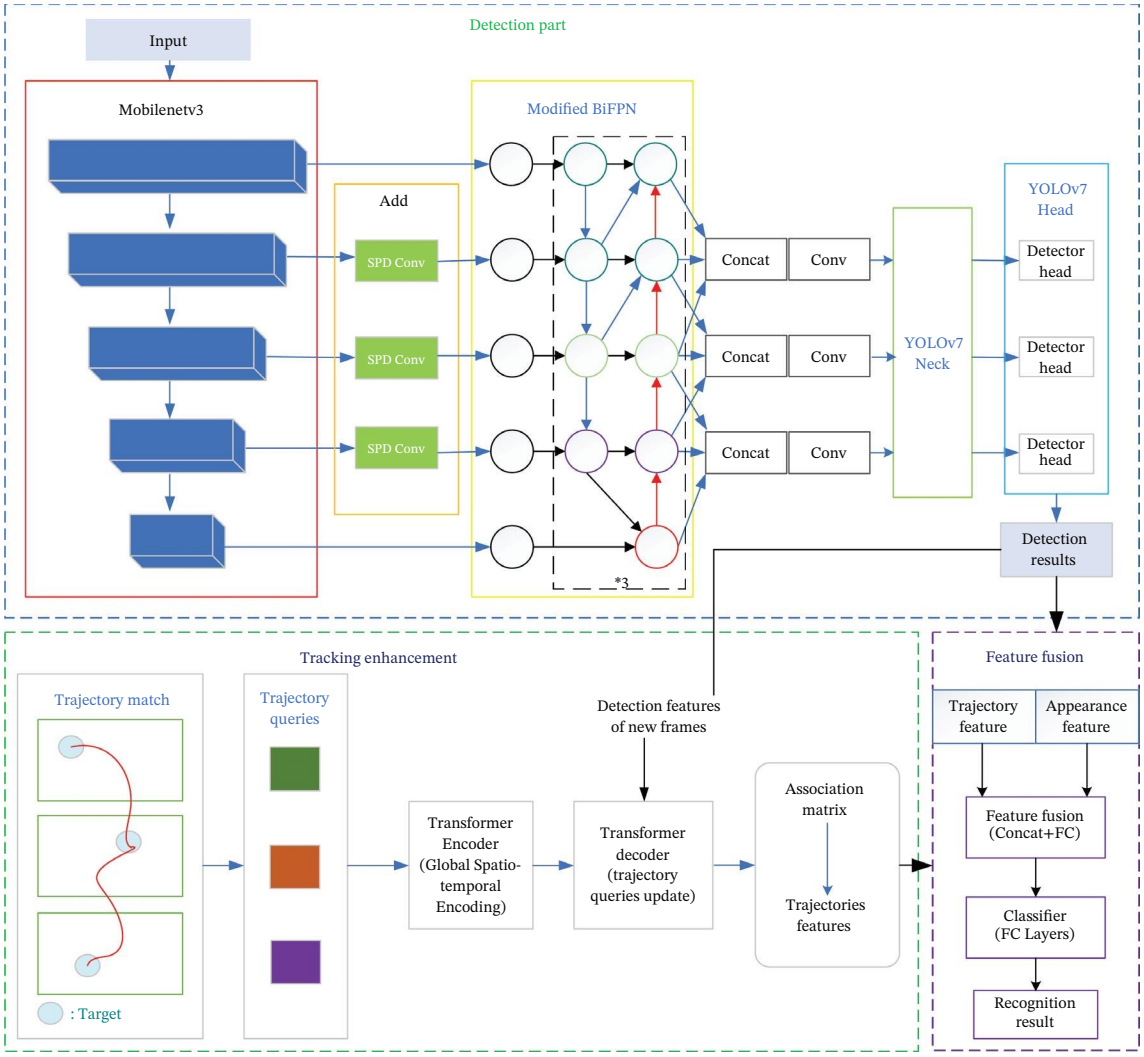



**FIGURE 6** | Overall architecture of the proposed UAV–bird recognition framework.

facilitate deployment on resource-constrained platforms, the original YOLOv7 backbone is replaced with MobileNetV3, which utilizes depthwise separable convolutions and lightweight bottleneck structures for efficient feature extraction. To improve the detection of small aerial targets, a small-object enhancement module combining a modified BiFPN and SPD-Conv is introduced into the neck network. The modified BiFPN strengthens multiscale feature fusion, while SPD-Conv preserves fine-grained spatial information that is often lost during conventional down-sampling operations. The enhanced features are then fed into the detection head to generate bounding boxes and preliminary target categories.

To further improve UAV–bird discrimination, the detection results from consecutive frames are forwarded to a temporal tracking branch. A simplified GTR is employed to associate targets across frames and extract trajectory-level motion features. By modeling long-term temporal dependencies, the tracking branch captures motion patterns that cannot be obtained from single-frame appearance information.



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The extracted trajectory information is fused with spatial detection features to perform target recognition. Because UAVs generally exhibit smoother and more stable trajectories, whereas birds often present periodic oscillations caused by wing-flapping motions, temporal motion characteristics provide valuable complementary information for classification. Through the joint utilization of appearance and motion cues, the proposed framework achieves robust UAV–bird recognition in complex low-altitude surveillance scenarios.

## 4 | Experiment Study

To evaluate the effectiveness of the constructed algorithm, it is necessary to conduct actual experiments and verify them by combining the experimental results with certain evaluation indicators. By comparing the experimental results of different algorithms, the actual value of this study is evaluated.

### 4.1 | Methods of Training and Testing

In the initial stages of the experiment, the YOLOv7 network is utilized with existing open-source weights for pretraining the dataset. However, due to GPU limitations, the batch size for training operations is set to eight. The batch size has a significant impact on the training speed of the network. Besides datasets for training, we also think about the validity of the testing model. We test the training results using some open-source datasets to verify the effectiveness of the model for relevant target recognition. The test part of MAV-VID [48] is used to validate the efficiency of the target tracking module of our model and Anti-UAV [49] dataset can be used to test the recognition effectiveness of our model for drones. In this study, SSD [6], YOLOv7 [8], YOLOv7-tiny, YOLOv9, YOLOv12, and YOLO26 models were employed to compare with proposed algorithms more intuitively. It is necessary to analyze the advantages and disadvantages of this algorithm in detection and recognition. This can make the algorithm training results have a stronger degree of credibility and can also provide a reliable reference for subsequent related research. To ensure the independence of the results, pretrained weights are not utilized. Instead, the model is initially trained on a private dataset. Subsequently, the trained model is tested on public datasets to ensure confidence in the results and facilitate comparisons with other methods.

The dataset is employed to train the algorithm model, resulting in the generation of a weight file specifically designed for the identification of drones and birds. Subsequently, this weight file is utilized during the evaluation phase by incorporating it into the test network. The test network evaluates the performance of the model on the test dataset. Throughout the training and testing process, the following results are obtained, reflecting the progression of steps: Training, the algorithm model is trained on the dataset, leveraging the obtained weight file. During this phase, the model learns to identify and classify drones and birds based on the provided training data. Testing, the trained model, equipped with the weight file, is then evaluated using the test dataset. The test network applies the model to the test dataset, analyzing its performance in identifying and distinguishing drones and birds.

As the training and testing progress, specific results are obtained, which reflect the model's effectiveness and accuracy in the identification task. These results provide insights into the performance of the algorithm and serve as a basis for evaluating its success in correctly identifying drones and birds within the test dataset.

All datasets were divided into mutually exclusive training, validation, and testing subsets, and no image overlap existed between training and testing sets.

### 4.2 | Experiment Environment

The study utilized the Python programming language and PyTorch [50] framework and using PyCharm to construct the structure of training algorithm based on open-source resources. All work took place on a normal personal computer with Intel CPU and Nvidia GPU. Nvidia RTX3080 GPU undertook the task of training and validation. Ubuntu is used as the system for all operations.

### 4.3 | Datasets Preparation

In recent years, an increasing number of datasets have been developed for object detection tasks. However, compared with common object categories such as pedestrians, vehicles, and animals, publicly available datasets specifically designed for UAV and bird recognition remain relatively limited. Existing UAV-related datasets are generally constructed for either UAV detection or object tracking tasks, and only a few datasets simultaneously consider UAV–bird discrimination under realistic low-altitude surveillance scenarios. Table 1 summarizes several representative datasets related to aerial target detection and recognition.

Each dataset listed in the table possesses unique characteristics regarding target type, scene complexity, and data modality. For example, Anti-UAV2021 mainly focuses on UAV detection and tracking under long-range surveillance conditions, whereas MAV-VID emphasizes moving aerial target tracking in video sequences. Other image-based datasets, such as USC-Drone and Det-Fly, contain a large number of

**TABLE 1** | Comparison of related datasets.

| Datasets | Type and quantity |
|---|---|
| Purdue [13] | 50 videos |
| Flying objects [14] | 14 videos |
| USC-drone [15] | 18778 pictures |
| Bird vs. Drone Detection Challenge [24] | 77 videos |
| Drone detection [51] | 114 videos |
| Real world [52] | 56821 pictures |
| MIDGARD [53] | 8775 pictures |
| MAV-VID [48] | 64 videos |
| Det-Fly [54] | 13271 pictures |
| Anti-UAV2021 [49] | 280 videos |



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UAV images but provide limited information regarding bird interference. Consequently, although these datasets are valuable for UAV detection research, they do not fully reflect the practical requirements of low-altitude surveillance systems, where UAVs and birds frequently coexist and exhibit highly similar visual appearances.

To better address the UAV–bird discrimination problem, a dedicated self-built UAV–Bird dataset was constructed in this study. The proposed dataset differs from existing public datasets in three aspects. First, both UAV and flying bird categories are explicitly included, enabling direct investigation of target confusion between these two visually similar aerial objects. Second, the dataset contains diverse target scales, ranging from nearby large targets to distant microtargets, which better represent practical surveillance conditions. Third, the images were collected under various illumination conditions, weather conditions, background complexities, and observation distances, thereby increasing the diversity of target appearances.

The dataset was created by combining self-collected aerial images with carefully selected open-source UAV and bird images. To improve robustness and reduce overfitting, several data augmentation strategies were applied, including noise injection and linear image transformation methods [55]. Most images were acquired at observation distances ranging from approximately 3 to 300 m, which closely corresponds to the operating range of typical low-altitude optical surveillance systems. As a result, the dataset better reflects real-world deployment scenarios than many existing benchmark datasets.

The final dataset contains 3998 annotated images with varying resolutions. Although its scale is smaller than some large-scale object detection benchmarks, it was specifically designed for UAV–bird recognition and includes a high proportion of challenging samples involving small targets, motion blur, bird interference, and complex backgrounds. To further evaluate the generalization capability of the proposed framework and reduce potential dataset bias, models trained on the self-built dataset were additionally evaluated on multiple public datasets, including MAV-VID, Anti-UAV RGB, and open source Drone vs. Birds Dataset from Kaggle. This cross-dataset validation strategy provides a more comprehensive assessment of the robustness and practical applicability of the proposed method.

Figure 7 presents several representative samples from the self-built dataset. Figure 8 illustrates the statistical distribution of target annotations, including category frequency, bounding-box scale distribution, and spatial location distribution. The dataset partitioning strategy used in this study is summarized in Table 2.

## 4.4 | Evaluation Metrics

In the evaluation of deep learning algorithms, there are many evaluation indicators. Table 3 shows some usual indexes which can judge performance of detection. The following four parameters contain all possible outcomes in target identification, and subsequent evaluation indicators are mainly calculated from them.

To be able to objectively evaluate this study, we selected some parameters to analyze the experimental results of this study, which are as follows:

### 4.4.1 | Mean Accuracy

Accuracy is the proportion of the total number of correctly predicted quantities in the prediction sample. The formula of accuracy is as follows:

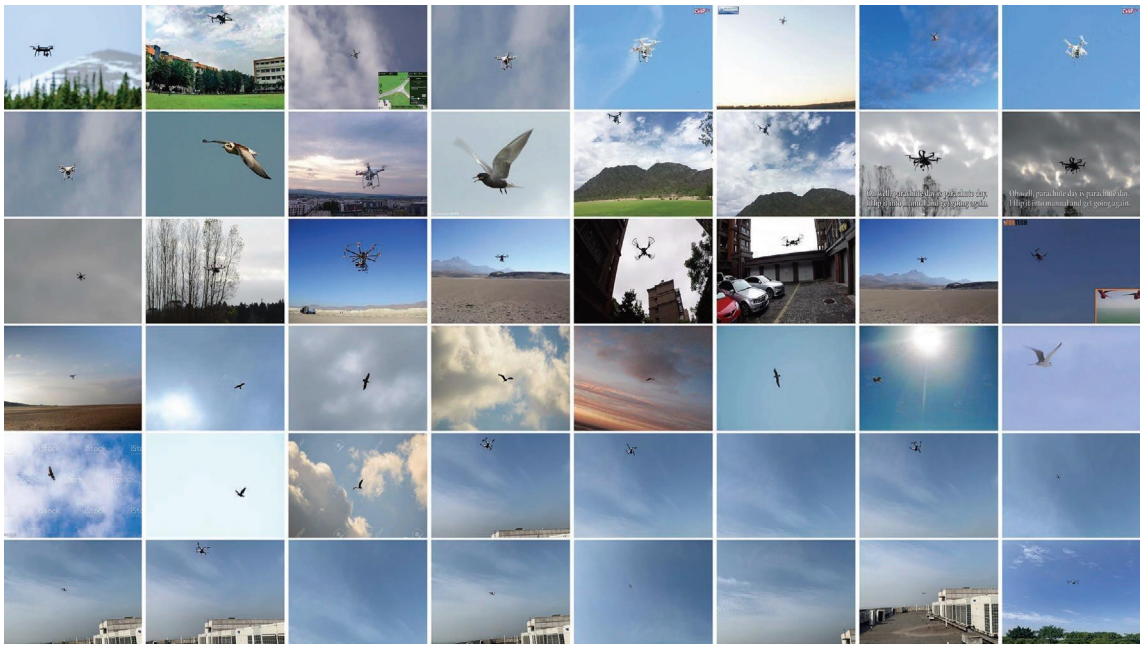


**FIGURE 7** | Representative samples from the self-built UAV–Bird dataset.



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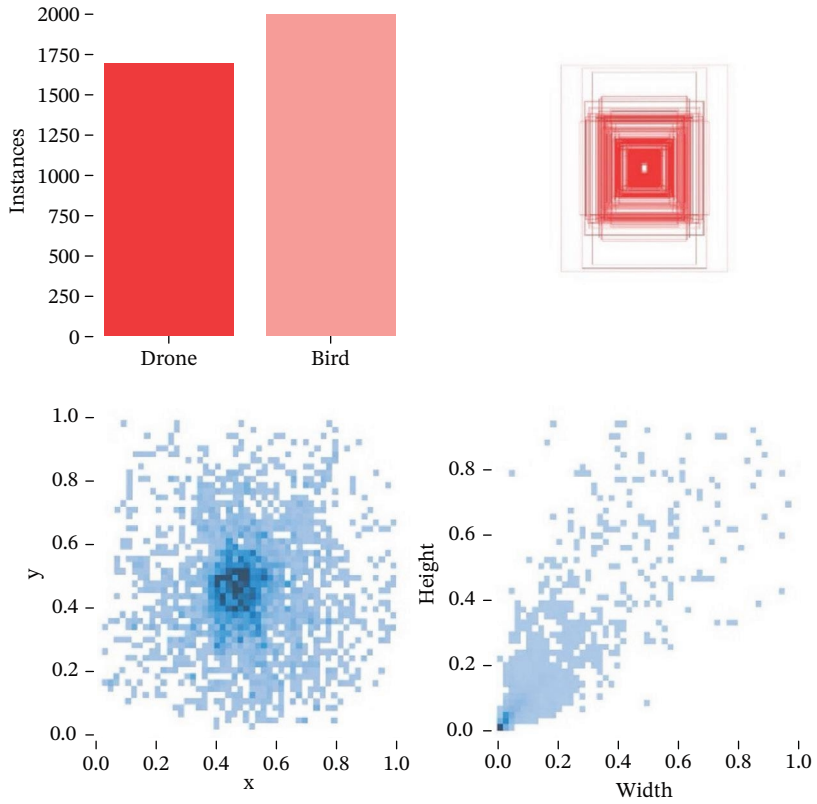



**FIGURE 8** | Statistics of image annotation data in the dataset.

**TABLE 2** | Dataset Partitioning.

| Dataset | Train | Val | Test | Total |
|---|---|---|---|---|
| **Number** | 2878 | 720 | 400 | 3998 |

**TABLE 3** | Confusion table, it covers all four situations that may occur in target recognition and expresses them with specific parameters.

| **Actual** | | |
|---|---|---|
| **Predict** | **True** | **False** |
| Positive | True positive (TP) | False positive (FP) |
| Negative | True negative (TN) | False negative (FN) |

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \tag{4}$$

Mean accuracy is the average of all category accuracy. It can more effectively display the overall accuracy of all categories, when a class of accuracy is too low, it will have a greater impact on the mean accuracy, only when the overall is more stable, the mean accuracy also shows a higher value. It is obtained by calculating the average of the sum of each category accuracy.

$$mAcc = \frac{\sum_{k=1}^{n} Accuracy(k)}{n}. \tag{5}$$

$n$ is number of all classes; $k$ is the number of a class of targets in the category.

#### 4.4.2 | Mean Average Recall (mAR)

Recall indicates the proportion of the actual positive sample size in the positive sample to the proportion of the whole sample in the positive sample. It represents the model's ability to find a target in an image. The formula of recall is as follows:

$$recall = \frac{TP}{TP + FN}. \tag{6}$$

Average recall is the result of averaging the recall values calculated separately for each class of target.

#### 4.4.3 | Mean Average Precision (mAP)

mAP is a widely used metric for evaluating target recognition performance. It calculates the mean of all average precisions. The formula of mAP is described as follows:

$$mAP = \frac{\sum_{q=1}^{Q} AveP(q)}{Q}. \tag{7}$$

$Q$ is a class in targets, $q$ indicates the number of kinds for all targets. $AveP(q)$ represents the average accuracy of $q$-type targets.

mAP considers the relationship between precision and recall in the recognition dynamically. It realizes the comprehensive evaluation of the results of all categories and uniformly evaluates the results of the algorithm in all categories, which can



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comprehensively and intuitively describe the advantages and disadvantages of the algorithm.

#### 4.4.4 | F1 Score

F1 scores are often used as the ultimate means of assessing the merits of a model. It is the harmonic average of precision and recall. The formula for F1 score is as follows:

$$F1 = 2 \bullet \frac{precision \bullet recall}{precision + recall}. \tag{8}$$

The formula of precision is as follows:

$$Percision = \frac{TP}{TP + FP}. \tag{9}$$

F1 score can also be described as follows:

$$F1 = 2 \bullet \frac{TP}{2 \bullet TP + FN + FP}. \tag{10}$$

The value of F1 score is between 0 and 1, and the closer its value is to 1, the better the model proves.

#### 4.4.5 | Frames Per Second (FPS)

FPS is a common way to measure the speed of processing action. It indicates how many images the algorithm can process in 1 s, which reflects the operation rate of the algorithm under certain conditions in disguise. In the computing task that has a certain time limit, the larger the value of FPS, the better.

## 5 | Results and Discussions

After the experiment, the results corresponding to the model and the dataset are obtained, and the corresponding results are analyzed and discussed to a certain extent.

### 5.1 | Experiment Results

After completing the training and validation process of the model, the corresponding results are obtained. These results represent the outcome of the algorithm's performance in identifying drones and birds. The results can include various metrics and evaluations, such as mAP recall, F1 score, and FPS.

Besides, we verify the effectiveness of each improved module in the actual application of the algorithm and obtain specific results. The actual improvement abilities of each module can be seen clearly.

### 5.2 | Discussions

The experimental results demonstrate that the proposed framework achieves a favorable balance between detection accuracy, computational efficiency, and practical deployment capability for low-altitude UAV–bird recognition tasks. Compared with conventional detectors, including SSD512-MobileNetV3, SSD512-VGG16, YOLOv7, and YOLOv7-tiny, the proposed method consistently achieves competitive performance across multiple public datasets and the self-built UAV–Bird dataset.

From Table 4, it can be observed that the proposed framework achieves the highest mAP@0.5:0.95 on both the Anti-UAV RGB dataset and the self-built UAV–Bird dataset while maintaining substantially lower computational complexity than the original YOLOv7 architecture. Although YOLOv7 obtains slightly higher recall values in some scenarios, its larger backbone network introduces significantly higher computational overhead, which limits real-time deployment on edge devices. In contrast, the proposed framework preserves most of the detection capability while reducing model complexity through the lightweight MobileNetV3 backbone; Table 5 compares the inference speed of different detection frameworks. The proposed method achieves 55.6 FPS, demonstrating an effective balance between detection performance and real-time deployment capability.

The comparison with recent lightweight detectors further highlights the effectiveness of the proposed design. YOLOv9n and YOLO26 achieve competitive detection accuracy owing to their advanced feature extraction strategies. However, these methods primarily focus on appearance-based object representation and do not explicitly exploit temporal motion information. In low-altitude surveillance scenarios, UAVs and birds often exhibit highly similar visual appearances, especially when targets are distant, blurred, or partially occluded. Under such conditions, single-frame detection approaches inevitably suffer from target confusion. By incorporating trajectory-aware recognition based on GTR, the proposed framework introduces temporal motion constraints into the recognition process, thereby improving discrimination capability between UAVs and birds.

Another important observation is that dataset characteristics significantly influence model performance. Models trained exclusively on Anti-UAV RGB may suffer from reduced robustness when deployed in practical surveillance scenarios containing complex background clutter. Similarly, MAV-VID-trained models demonstrate strong localization performance but occasionally fail to maintain continuous target tracking in long-duration sequences. In contrast, the self-built dataset contains diverse bird interference samples, multiscale targets, varying illumination conditions, and realistic surveillance distances, making it more representative of practical low-altitude monitoring scenarios. Consequently, models trained on the self-built dataset exhibit superior robustness and generalization capability during real-world testing.

To further analyze the optimization behavior of the proposed framework, the training curves and precision–recall (P–R) curves are illustrated in Figure 9. As shown in Figure 9a, the box loss, objectness loss, and classification loss continuously decrease during training and gradually converge after approximately 200 epochs. Similar convergence trends can also be observed in the validation losses, indicating that the proposed framework achieves stable optimization without obvious overfitting. Meanwhile, precision, recall, mAP@0.5, and



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**TABLE 4** | Results comparisons between different methods; we use four evaluation metrics to evaluate the results of training on multiple datasets using different algorithms.

| Datasets | Model | mAP@0.5 | mAP@0.5:0.95 | Recall | F1 score | Parameters/M | FLOPs/G |
|---|---|---|---|---|---|---|---|
| MAV-VID [48] | SSD512-MobileNetV3 | 0.894 | 0.574 | 0.623 | 0.719 | 5.4 | 1.5 |
| | SSD512-VGG16 | 0.909 | 0.535 | 0.612 | 0.742 | 37.1 | 99.5 |
| | YOLOv7 | 0.997 | 0.748 | 0.997 | 0.996 | 37.2 | 104.7 |
| | YOLOv7-tiny | 0.996 | 0.733 | 0.990 | 0.994 | 6.2 | 13.1 |
| | YOLOv9n | 0.995 | 0.752 | 0.997 | 0.997 | 2.9 | 7.7 |
| | YOLOv12n | 0.995 | 0.749 | 0.997 | 0.997 | 2.6 | 6.5 |
| | YOLO26 | 0.995 | 0.756 | 0.992 | 0.991 | 2.4 | 5.2 |
| | Our method | 0.997 | 0.733 | 0.991 | 0.995 | 26.3 | 66.8 |
| Anti-UAV RGB [49] | SSD512-MobileNetV3 | 0.536 | 0.401 | 0.498 | 0.529 | 5.4 | 1.5 |
| | SSD512-VGG16 | 0.979 | 0.627 | 0.703 | 0.780 | 37.1 | 99.5 |
| | YOLOv7 | 0.997 | 0.733 | 0.993 | 0.995 | 37.2 | 104.7 |
| | YOLOv7-tiny | 0.997 | 0.721 | 0.993 | 0.995 | 6.2 | 13.1 |
| | YOLOv9n | 0.995 | 0.789 | 0.995 | 0.997 | 2.9 | 7.7 |
| | YOLOv12n | 0.995 | 0.796 | 0.996 | 0.997 | 2.6 | 6.5 |
| | YOLO26 | 0.995 | 0.816 | 0.995 | 0.996 | 2.4 | 5.2 |
| | Our method | 0.998 | 0.811 | 0.995 | 0.996 | 26.3 | 66.8 |
| Drone vs. Birds dataset [56] | SSD512-MobileNetV3 | 0.620 | 0.397 | 0.526 | 0.496 | 5.4 | 1.5 |
| | SSD512-VGG16 | 0.894 | 0.548 | 0.564 | 0.627 | 37.1 | 99.5 |
| | YOLOv7 | 0.913 | 0.489 | 0.860 | 0.896 | 37.2 | 104.7 |
| | YOLOv7-tiny | 0.840 | 0.415 | 0.819 | 0.851 | 6.2 | 13.1 |
| | YOLOv9n | 0.946 | 0.522 | 0.933 | 0.937 | 2.9 | 7.7 |
| | YOLOv12n | 0.921 | 0.510 | 0.899 | 0.914 | 2.6 | 6.5 |
| | YOLO26 | 0.933 | 0.518 | 0.903 | 0.915 | 2.4 | 5.2 |
| | Our method | 0.895 | 0.514 | 0.813 | 0.874 | 26.3 | 66.8 |
| Self-built UAV–Bird dataset | SSD512-MobileNetV3 | 0.530 | 0.392 | 0.455 | 0.478 | 5.4 | 1.5 |
| | SSD512-VGG16 | 0.844 | 0.482 | 0.573 | 0.619 | 37.1 | 99.5 |
| | YOLOv7 | 0.927 | 0.577 | 0.901 | 0.903 | 37.2 | 104.7 |
| | YOLOv7-tiny | 0.906 | 0.575 | 0.866 | 0.896 | 6.2 | 13.1 |
| | YOLOv9n | 0.914 | 0.572 | 0.893 | 0.920 | 2.9 | 7.7 |
| | YOLOv12n | 0.906 | 0.561 | 0.879 | 0.909 | 2.6 | 6.5 |
| | YOLO26 | 0.935 | 0.584 | 0.901 | 0.925 | 2.4 | 5.2 |
| | Our method | 0.933 | 0.596 | 0.894 | 0.914 | 26.5 | 66.8 |



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**TABLE 5** | Average FPS in different algorithms.

| Model | SSD512-MobileNetV3 | SSD512-VGG16 | YOLOv7 | YOLOv7-tiny | YOLOv9n | YOLOv12n | YOLO26 | Our method |
|---|---|---|---|---|---|---|---|---|
| Average FPS | 39.3 | 30.2 | 33.6 | 144.3 | 274.1 (RTX4060) | 377.3 (RTX4060) | 769 (RTX4060) | 55.6 |

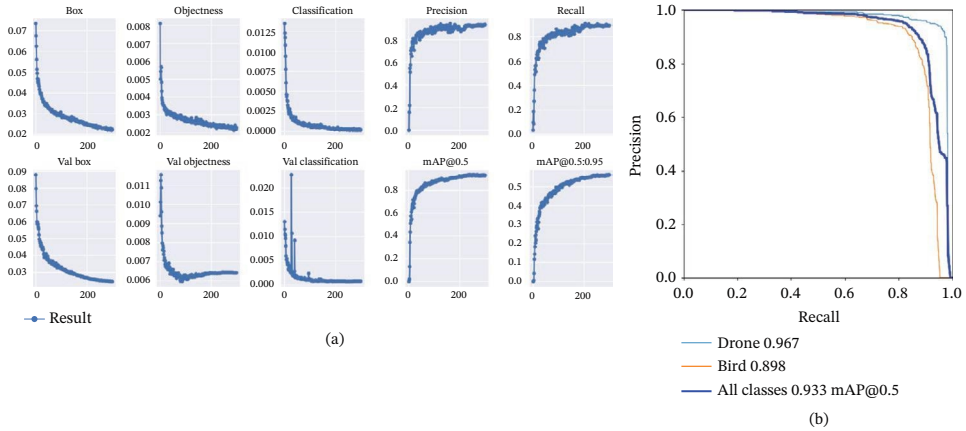



**FIGURE 9** | Training convergence curves and precision–recall curves of the proposed method. (a) Training and validation losses, precision, recall, and mAP metrics during training. (b) Precision–recall curves for UAV and bird categories.

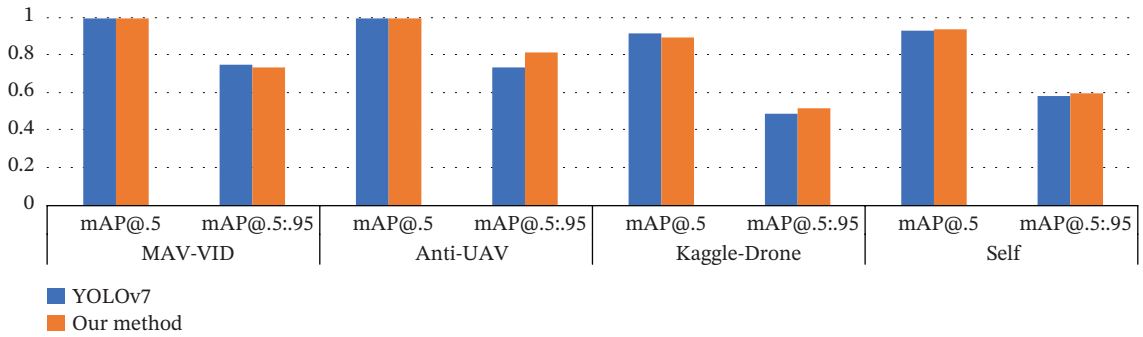



**FIGURE 10** | Comparisons between YOLOv7 and our method on different datasets.

mAP@0.5:0.95 consistently increase throughout the training process and eventually stabilize, demonstrating the effectiveness of the proposed feature extraction and fusion strategies.

Figure 9b presents the P–R curves for different categories. The UAV category achieves an average precision of 0.967, whereas the bird category achieves 0.898. The overall mAP@0.5 reaches 0.933, indicating that the proposed framework maintains strong discriminative capability for both UAVs and birds. The slightly lower performance on bird targets can be attributed to their larger appearance variations, motion blur, and irregular flight patterns in practical surveillance scenarios. Nevertheless, the proposed method still achieves high precision and recall across both categories, confirming its robustness in low-altitude aerial target recognition tasks.

As shown in Figure 10, the proposed method produces more accurate bounding boxes and fewer false detections than the original YOLOv7, particularly for distant UAVs and small bird targets. Compared with the original YOLOv7, the proposed method demonstrates more stable detection performance across different target scales. Improvements are particularly evident for small UAVs and distant flying birds, where conventional down-sampling operations tend to eliminate critical spatial information. The integration of modified BiFPN and SPD-Conv effectively preserves fine-grained features, resulting in improved localization accuracy for small-scale targets.

In addition to detection performance, computational efficiency remains a critical consideration for practical deployment. Although the introduction of trajectory modeling inevitably increases computational requirements, the lightweight backbone compensates for this overhead. As a result, the proposed framework achieves 55.6 FPS, significantly exceeding the real-time processing requirement for surveillance applications while maintaining competitive recognition performance. This balance between accuracy and efficiency makes the proposed framework particularly suitable for airport perimeter surveillance,



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**TABLE 6** | Ablation experiment results.

| Model | MAP@0.5 | MAP@0.5:0.95 | Recall | F1 score | FPS |
|---|---|---|---|---|---|
| YOLOv7 | 0.927 | 0.577 | 0.901 | 0.903 | 33.6 |
| +MobileNetV3-large | 0.883 | 0.519 | 0.834 | 0.864 | 67.8 |
| +Modified BiFPN | 0.931 | 0.563 | 0.890 | 0.912 | 32.8 |
| +SPD-Conv | 0.913 | 0.592 | 0.843 | 0.877 | 33.2 |
| +Global Tracking Transformer | 0.929 | 0.582 | 0.884 | 0.892 | 16.4 |
| +MobileNetV3 + modified BiFPN | 0.919 | 0.567 | 0.875 | 0.897 | 61.4 |
| +MobileNetV3 + SPD-Conv | 0.922 | 0.584 | 0.881 | 0.901 | 63.7 |
| +MobileNetV3 + modified BiFPN +SPD-Conv | 0.928 | 0.589 | 0.889 | 0.908 | 58.9 |
| Our method (all combined) | 0.933 | 0.596 | 0.894 | 0.914 | 55.6 |

low-altitude security monitoring, and intelligent airspace management systems.

### 5.3 | Ablation Study

To evaluate the contribution of each proposed module, ablation experiments were conducted using YOLOv7 as the baseline network. The results are presented in Table 6.

Replacing the original backbone with MobileNetV3 significantly improves inference speed from 33.6 to 67.8 FPS, demonstrating the effectiveness of the lightweight design. However, the reduction in model capacity leads to decreases in mAP and F1 score. In contrast, the modified BiFPN improves feature fusion across different scales, increasing mAP@0.5 from 0.927 to 0.931 and F1 score from 0.903 to 0.912. The SPD-Conv module further enhances localization performance by preserving spatial information during down-sampling, resulting in the highest mAP@0.5:0.95 (0.592) among all single-module configurations. Meanwhile, the GTR module introduces temporal motion information and slightly improves detection performance, although additional trajectory modeling reduces the inference speed.

For further investigation of module complementarity, several combined configurations were evaluated. The combination of MobileNetV3 and modified BiFPN effectively recovers the accuracy loss caused by lightweighting, whereas MobileNetV3 and SPD-Conv achieve a better balance between detection accuracy and computational efficiency. When MobileNetV3, modified BiFPN, and SPD-Conv are simultaneously integrated, the detector achieves performance comparable to the original YOLOv7 while maintaining substantially higher inference speed.

Finally, the complete framework integrating MobileNetV3, modified BiFPN, SPD-Conv, and GTR achieves the best overall performance, obtaining 0.933 mAP@0.5, 0.596 mAP@0.5:0.95, and an F1 score of 0.914 at 55.6 FPS. Compared with the baseline YOLOv7, the proposed method improves both detection accuracy and real-time performance, demonstrating that the proposed modules provide complementary benefits for low-altitude UAV and bird recognition.

## 6 | Conclusion

In this paper, an improved YOLOv7-based framework for low-altitude UAV and flying bird recognition is proposed. The framework is designed to address several practical challenges encountered in real-world aerial surveillance scenarios, including small target size, appearance similarity between UAVs and birds, complex backgrounds, and target motion uncertainty. To improve the applicability of the detector in resource-constrained environments, MobileNetV3 is adopted as a lightweight backbone to reduce computational complexity while maintaining satisfactory feature extraction capability. Furthermore, a modified BiFPN structure and SPD-Conv module are introduced to enhance multiscale feature fusion and preserve fine-grained spatial information, thereby improving the detection performance of small aerial targets. In addition, a trajectory-aware recognition strategy based on GTR is incorporated to exploit temporal motion characteristics for distinguishing UAVs from flying birds.

To evaluate the proposed framework, extensive experiments were conducted on multiple public datasets as well as a self-built UAV–Bird dataset. The experimental results demonstrate that the proposed method achieves competitive detection accuracy while maintaining real-time inference capability. Compared with the original YOLOv7, the proposed framework improves mAP@0.5 from 0.927 to 0.933 and F1 score from 0.903 to 0.914 while simultaneously increasing the inference speed from 33.6 to 55.6 FPS. The ablation studies further verify the effectiveness and complementarity of the proposed modules in balancing detection accuracy, robustness, and computational efficiency.

Self-built UAV–Bird dataset containing realistic low-altitude surveillance scenarios was constructed to support the training and evaluation of appearance- and motion-based recognition methods. The dataset includes diverse target scales, motion patterns, background conditions, and bird interference cases, providing valuable reference data for future UAV recognition research.

In total, the proposed framework provides an effective and practical solution for low-altitude UAV and bird recognition and demonstrates strong potential for deployment in airport



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surveillance, civil aviation safety management, and intelligent airspace monitoring systems. Future work will focus on improving cross-dataset generalization capability, introducing more efficient temporal modeling mechanisms, and extending the framework to multitarget collaborative tracking and recognition tasks in complex low-altitude environments.


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### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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